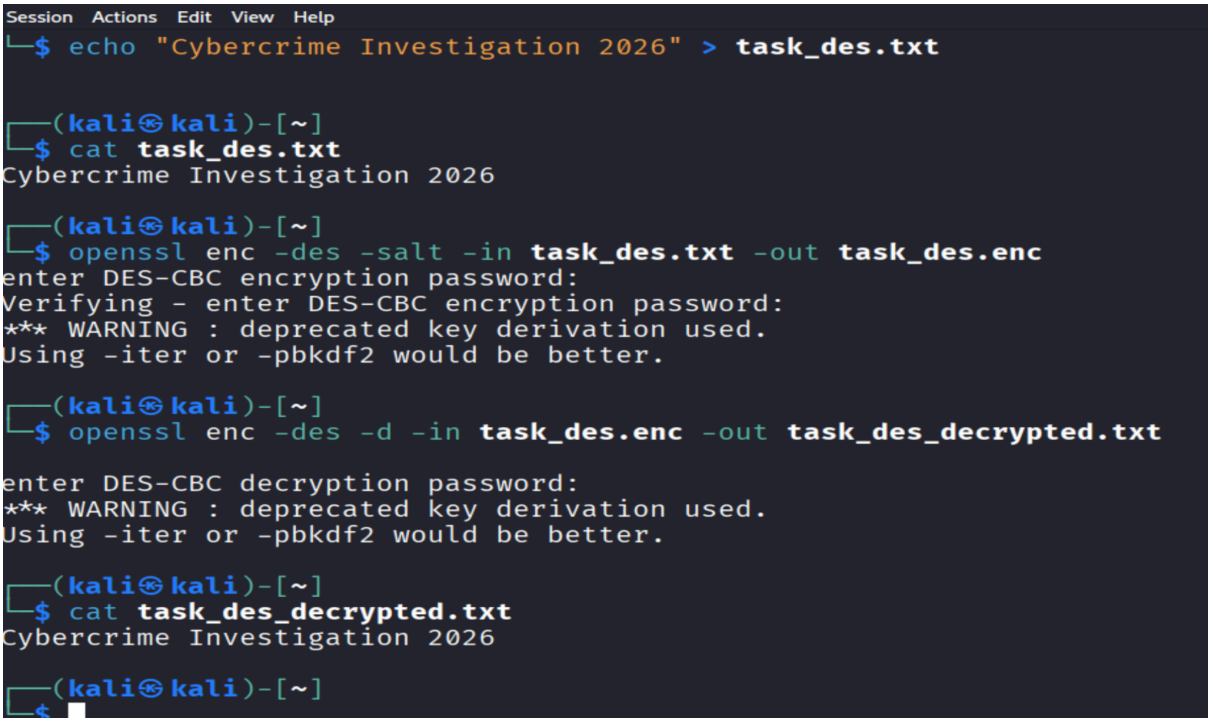
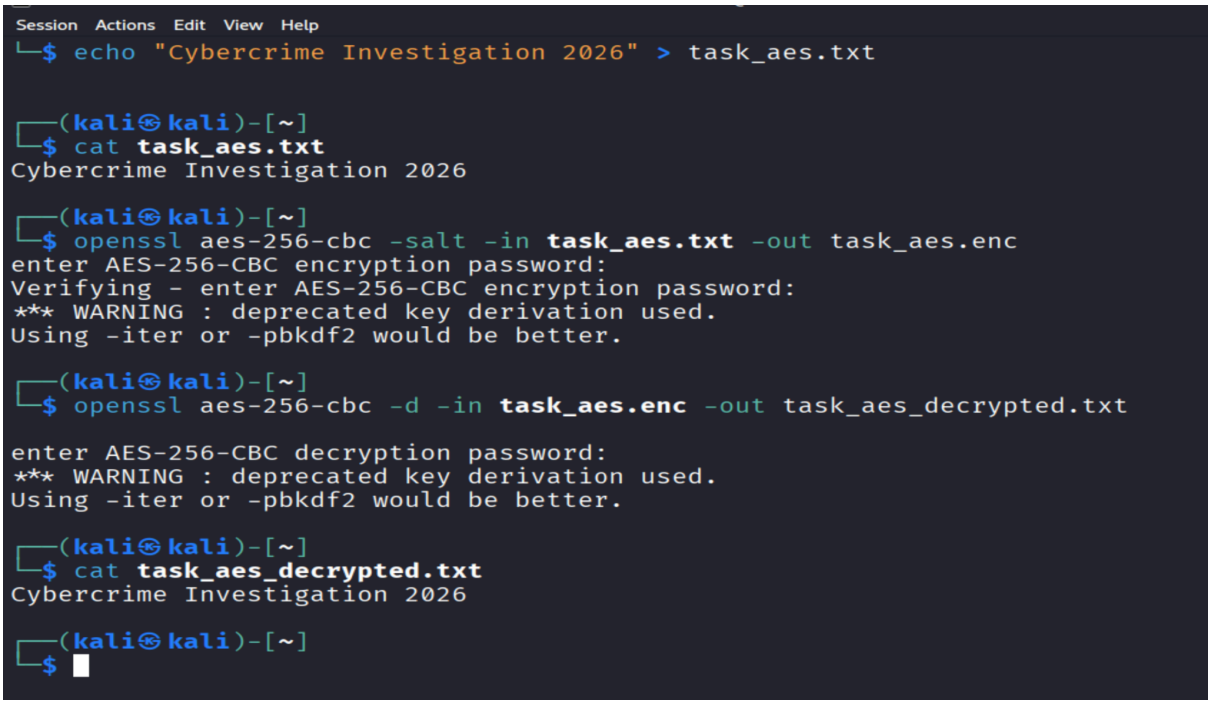
Task - DES vs AES

- DES : I encrypted and decrypted a text file using DES, and successfully verified that the original text was recovered.



- AES: I encrypted and decrypted a text file using AES-256, and successfully verified that the original text was recovered.



- Key Size: DES: 56-bit , AES:256-bit .

- Security : DES: Weak, breakable by brute force
  AES: Strong, much harder to break due to its large key size (up to 256-bit)

- Speed
- DES: Slower — no modern hardware acceleration
- AES: Faster — supported by AES-NI on modern CPUs

- Usage Today
- DES: Deprecated — officially withdrawn by NIST.
- AES: Global standard — HTTPS, VPNs, WhatsApp.

- Why is DES not secure today?
- DES only uses a 56-bit key, which makes it vulnerable to brute force attacks .

- Why is AES the better choice?
- AES uses a much larger key 256 bit making brute force attacks practically impossible .

- What happens if an attacker gets the encryption key?
- All encrypted data can be decrypted and read, and the attacker can also forge or tamper with data using that key.

- I chose AES-256, because it's more secure, faster, and the current global standard, while DES is outdated and breakable.

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